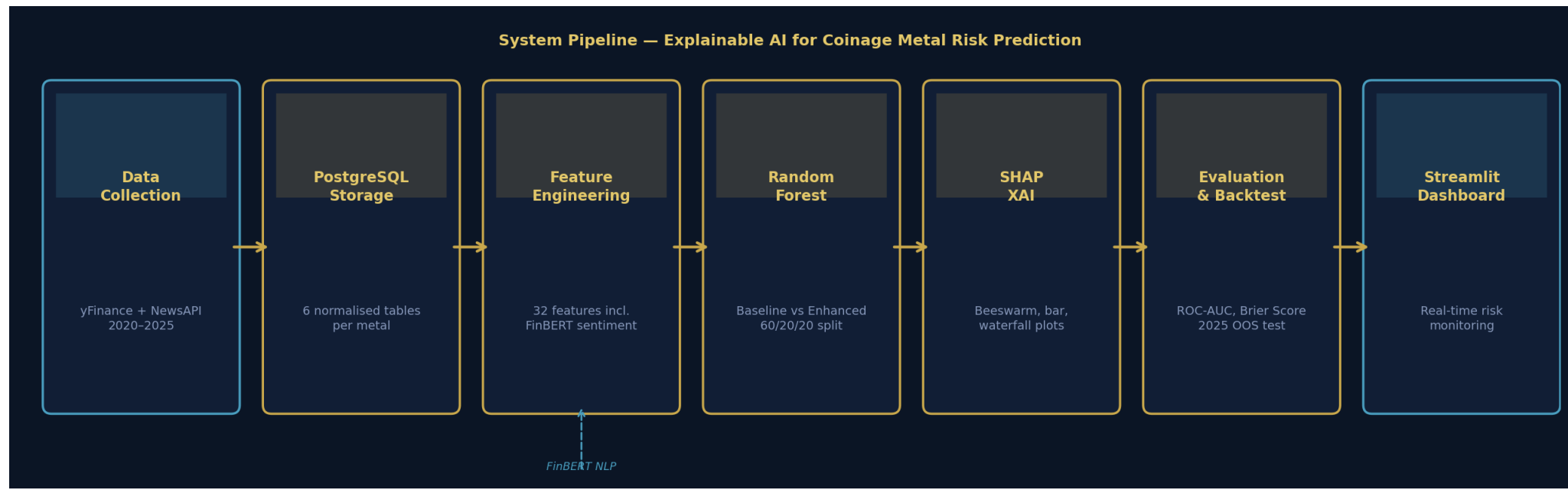




## Abstract

This project develops an explainable AI framework for predicting daily downside risk events which here is defined as price declines exceeding 2%, in gold, silver, and copper futures markets.

A Random Forest classifier is trained on technical, macroeconomic, and NLP-derived sentiment features and evaluated using probabilistic metrics (ROC-AUC, Brier Score) alongside SHAP explanations to ensure transparent, interpretable risk scores.

The enhanced sentiment model demonstrated improved calibration for gold and copper, with SHAP analysis revealing that macroeconomic drivers, particularly the 10-year Treasury yield, VIX, and S&P 500 returns, dominate risk prediction across all three metals.

Results are stored in a PostgreSQL database and visualized via a Streamlit dashboard supporting near real-time risk monitoring.

## Introduction

Coinage metals which are gold, silver, and copper are central to global economic activity, serving simultaneously as safe-haven assets, industrial inputs, and macroeconomic indicators. Their short-term prices are highly sensitive to geopolitical events, currency fluctuations, and shifting market sentiment.

Traditional econometric models (ARIMA, GARCH) rely on strict statistical assumptions that limit their ability to model the non-linear, dynamic volatility observed in commodity markets. While ML algorithms address this, they often function as opaque "black boxes," generating predictions without interpretable justification. This a critical problem in financial risk contexts where accountability is essential.

This project addresses two concurrent gaps in the literature: the lack of explainable AI applications in commodity markets, and the limited integration of NLP sentiment with interpretable models for metal risk prediction.

## Aims and Objectives

To develop an explainable ML model for predicting short-term downside risk in gold, silver, and copper markets by integrating technical indicators with NLP-derived sentiment, producing SHAP-backed probability risk scores.

### Objectives:

- Evaluate whether FinBERT sentiment features improve prediction over technical indicators alone
- Develop and validate a Random Forest classifier for daily downside risk (>2% decline)
- Assess model calibration using Brier Score, Log Loss, and calibration curves
- Determine which feature categories (technical, macro, sentiment, temporal) drive prediction
- Validate SHAP explanations against historical market events
- Deploy results via a Streamlit dashboard for real-time risk monitoring

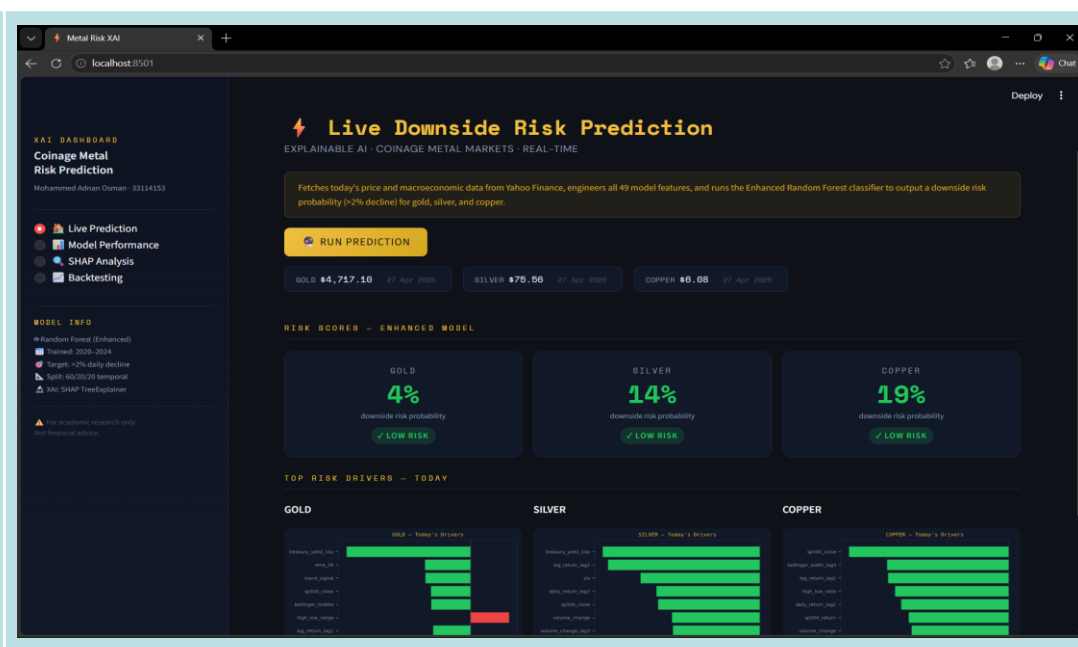
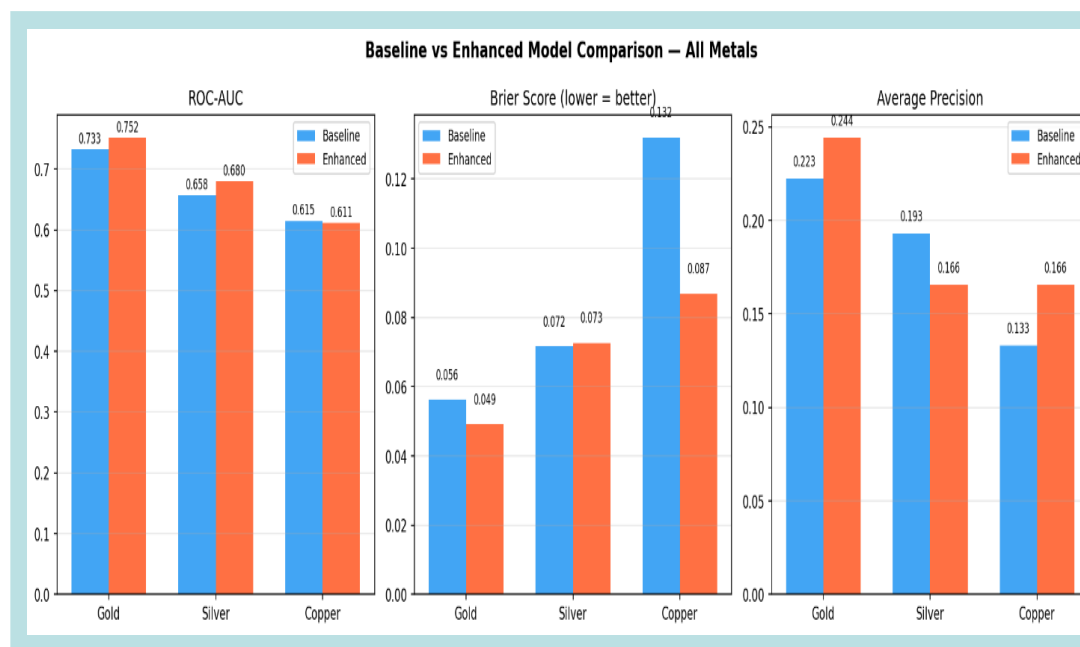
### Data Sources:

- Price data: yFinance API (GC=F, SI=F, HG=F) From 2020 -> 2025
- Macroeconomic: VIX, DXY, 10-year Treasury yield, S&P 500
- Sentiment: NewsAPI financial headlines processed via FinBERT
- Storage: PostgreSQL database (metal\_risk\_prediction)

**Research question:** Can a Random Forest classifier integrating FinBERT sentiment analysis accurately predict daily downside risk events (>2% price decline) in gold, silver, and copper markets, while providing interpretable probability-based risk scores and SHAP explanations that align with established financial risk drivers?

## Conclusions

- Copper showed greatest sentiment impact (Brier 0.132 -> 0.087, -34%); Silver worsened (AP 0.193 -> 0.166), confirming metal-specific FinBERT sensitivity.
- Gold strongest overall (AUC 0.752 test, 0.847 backtest); treasury\_yield\_10y dominates with inverse risk relationship.
- Sentiment absent from all SHAP top-20 rankings despite improving calibration. Latent non-linear ensemble contribution rather than direct effect.



Future work includes applying SMOTE or cost-sensitive learning for class imbalance, extending to LSTM/XGBoost for temporal dependency, productionising the Streamlit dashboard with automated retraining, and investigating cross-metal ratio features as additional predictive signals

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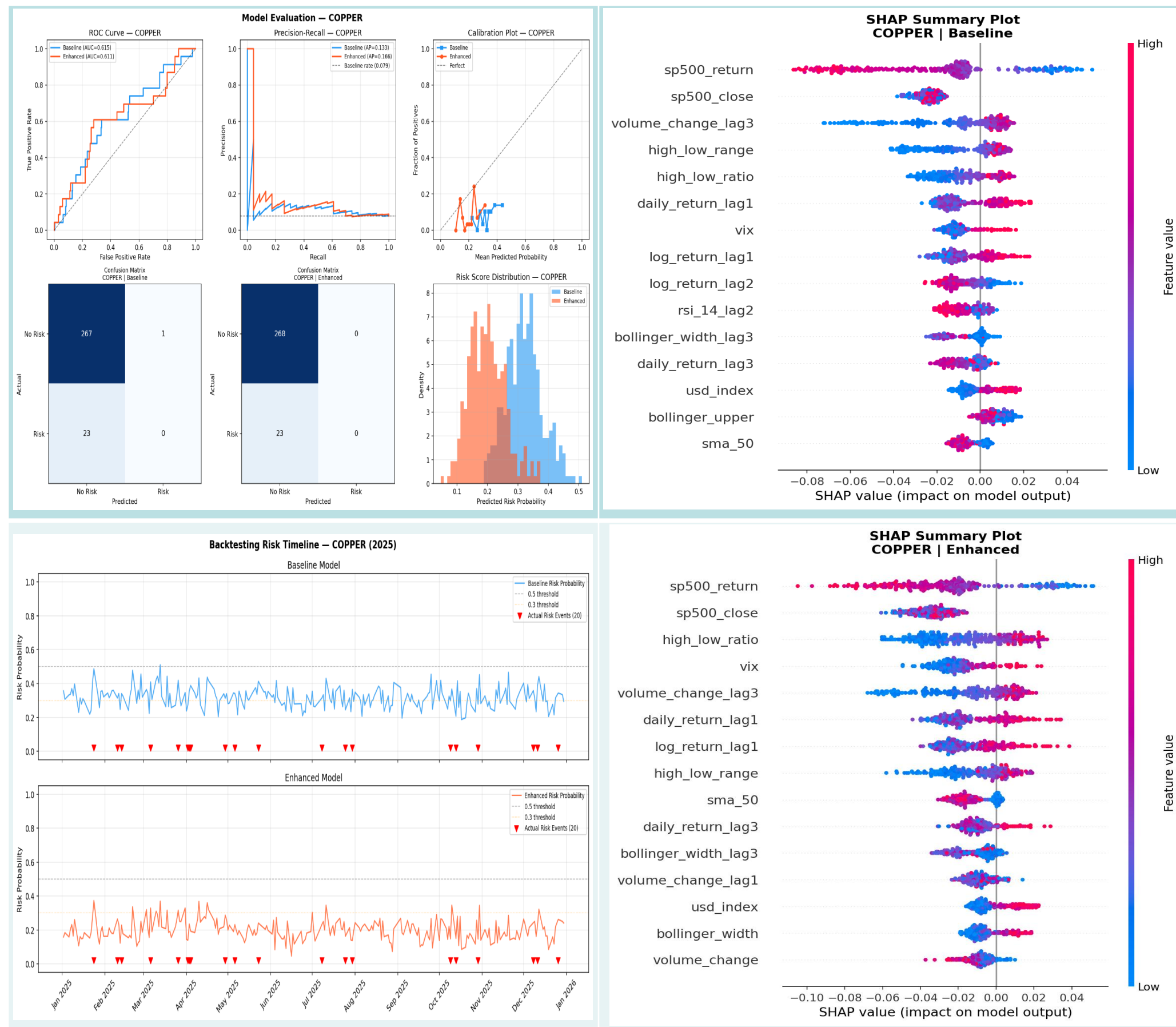
## Methodology

- Data Collection:** Historical OHLCV price data for gold (GC=F), silver (SI=F), and copper (HG=F) collected via yFinance API covering 2020–2025, alongside macroeconomic indicators (VIX, DXY, 10-year Treasury yield, S&P 500) and financial news headlines via NewsAPI.
- PostgreSQL Storage:** All collected data stored in a PostgreSQL database (metal\_risk\_prediction) with six normalised tables per metal. SQLAlchemy 2.0 ORM used for type-safe upserts, managed via pgAdmin.
- Feature Engineering:** 32 features engineered per metal including daily returns, log returns, Bollinger Bands, RSI, ATR, moving averages (5/10/20-day), high-low range, volume change, macroeconomic indicators, temporal flags, and FinBERT compound sentiment scores.
- Random Forest:** Binary classifier trained using a 60/20/20 train-validation-test split. GridSearchCV applied for hyperparameter optimisation. class\_weight='balanced' used to address 7.3% class imbalance. Two model variants trained per metal, the baseline and sentiment-enhanced.
- SHAP XAI:** SHAP TreeExplainer applied to both model variants producing beeswarm, bar, waterfall, force, and dependence plots. Top-20 feature rankings generated per metal to identify dominant risk drivers.
- Evaluation & Backtest:** Models assessed using ROC-AUC, Brier Score, Log Loss, Average Precision, and calibration curves. Out-of-sample 2025 backtesting conducted comparing risk-signal strategy against buy-and-hold benchmark.
- Streamlit Dashboard:** All predictions, risk scores, and SHAP explanations deployed via an interactive Streamlit dashboard supporting near real-time daily risk monitoring across all three metals.

## Results

Table 1: Full evaluation metrics for all six model variants on the held-out test set

| Metal  | Model    | ROC-AUC |        | Avg Precision |        | Brier Score |     | Log Loss | F1 |
|--------|----------|---------|--------|---------------|--------|-------------|-----|----------|----|
|        | N_test   | N_risk  |        |               |        |             |     |          |    |
| Gold   | Baseline | 0.7327  | 0.2225 | 0.0562        | 0.2341 | 0.0         | 290 | 16       |    |
| Gold   | Enhanced |         | 0.7516 | 0.2443        | 0.0491 | 0.1968      | 0.0 | 290      | 16 |
| Silver | Baseline | 0.6577  | 0.1931 | 0.0716        | 0.2799 | 0.0         | 278 | 20       |    |
| Silver | Enhanced |         | 0.6803 | 0.1657        | 0.0727 | 0.2841      | 0.0 | 278      | 20 |
| Copper | Baseline | 0.6147  | 0.1333 | 0.1319        | 0.4446 | 0.0         | 291 | 23       |    |
| Copper | Enhanced |         | 0.6111 | 0.1657        | 0.0868 | 0.3261      | 0.0 | 291      | 23 |



### EXAMPLE CASE STUDY METAL: COPPER

- F1 = 0 across all models is an expected consequence of 7.3% class imbalance in rare-event classification, not model failure
- NewsAPI free tier limited headline volume. Other broader sources may strengthen the NLP signal
- Models trained on historical data may struggle to generalise to unprecedented market conditions